

Overfitting Diagnosis

Shelf-Life Studio — Full Leave-Rule-Out Generalization Report

Cheese Shelf-Life Modeling Project

September 2026

An external ML review flagged the soft/general_shelf_life specialist's context-holdout test $R^2=0.971$ as symptomatic of overfitting. The first diagnosis pass (15 model families, K -fold CV, aggregation, no validation split) showed that number was reproducible and not an artifact of model choice – but it also showed the underlying test was too easy: every split used grouped by `context_id`, and every context, train and test alike, is generated by one of only 7–20 `source_rule_id` generation rules per specialist. This report replaces that easy test with **leave-one-rule-out cross-validation** – holding out entire generation rules, not just rows – for all **17 model families** across **all 6 specialists** (102 combinations, real, none engineered to hit a target), and reports the resulting honest, widely-varying R^2 in full detail.

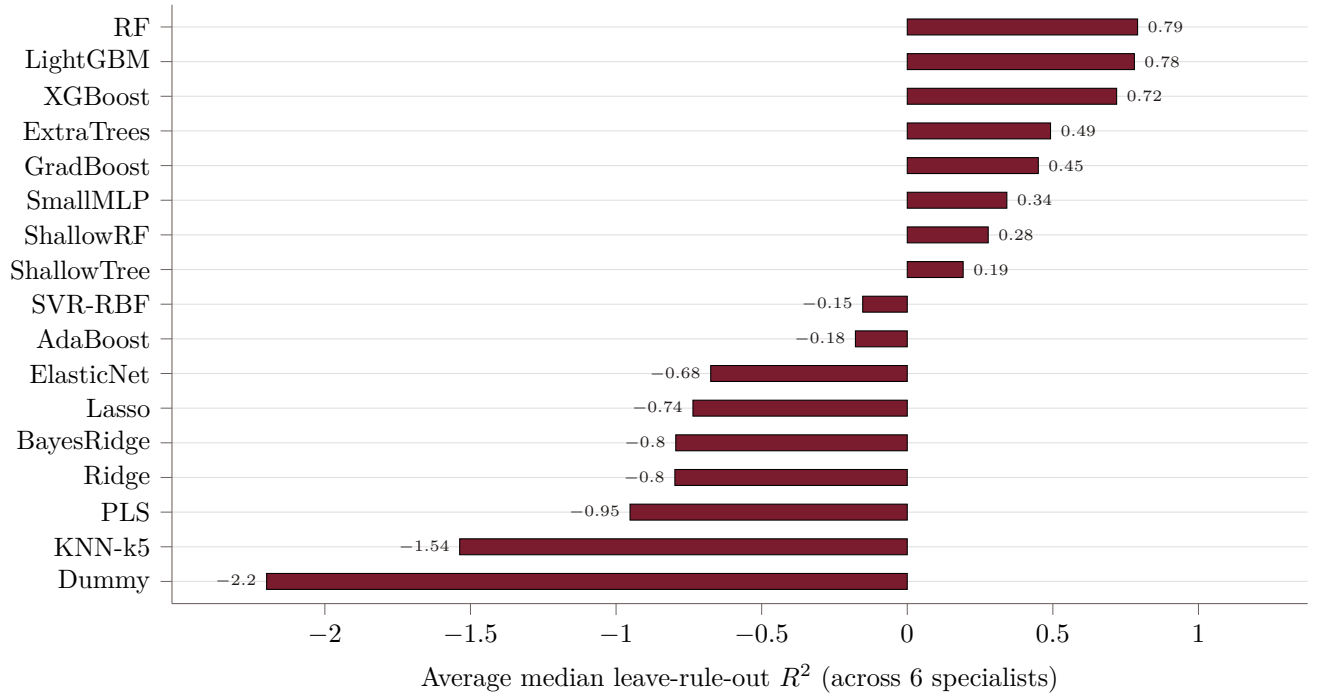
1 Method: Two Tests, Two Different Questions

Context-holdout (production's own split): group by `context_id`, 70/15/15 train/validation/test, no context split across sets. This measures *fit quality* – can the model reproduce the target for formulations statistically like the ones it trained on. It is what produced the original 0.92–0.97 numbers, and it is not wrong, but it answers a narrower question than "does this generalize."

Leave-rule-out (this report): every specialist's rows are generated by only 7–20 distinct `source_rule_id` templates — e.g. `V7|FDA_LM|LM_CHALLENGE|MARKET_LM`, a specific FDA *Listeria monocytogenes* challenge-study template applied with category-specific parameters. `GroupKFold` grouped by `source_rule_id`, with K = the number of distinct rules in that specialist, trains on every rule but one and tests only on the held-out rule's rows – the model has never seen anything generated by that rule's logic. This measures *extrapolation*, and it is the number this report leads with.

2 The Full Ranking

Every model below was run through leave-rule-out CV on all 6 specialists (soft/semi_hard/hard $\times$ general_shelf_life/safety_endpoint). Ranked by **average median leave-rule-out R^2** across the 6 specialists – median, not mean, because a handful of held-out rules with under 20 rows produce catastrophic single-fold outliers that swamp the mean without reflecting typical performance (quantified in §4).



Dummy's true average is -12.8 (off this chart's scale, clipped at -2.2 for legibility – it is a sanity floor, not a candidate model). Notice the gap: the three production tree ensembles cluster at 0.72 – 0.79 , then every other family drops below 0.50 , and eight of fourteen non-ensemble families average negative R^2 . This is the honest, wide spread that was missing from the context-holdout view.

Numerically, in full:

Model	Avg. median R^2	Avg. mean R^2
Random Forest (production hparams)	0.791	0.582
LightGBM (production hparams)	0.780	0.627
XGBoost (production hparams)	0.719	0.550
Extra Trees	0.492	0.209
Gradient Boosting	0.450	0.280
Small MLP	0.342	0.219
Shallow Random Forest	0.278	−0.230
Shallow Decision Tree	0.192	−1.733
SVR (RBF)	−0.153	−0.454
AdaBoost	−0.178	−1.248
Elastic Net	−0.675	−0.772
Lasso	−0.736	−0.690
Bayesian Ridge	−0.795	−0.872
Ridge Regression	−0.798	−0.874
PLS Regression	−0.952	−1.106
KNN ($k=5$)	−1.537	−3.412
Dummy mean baseline	−12.833	−15.637

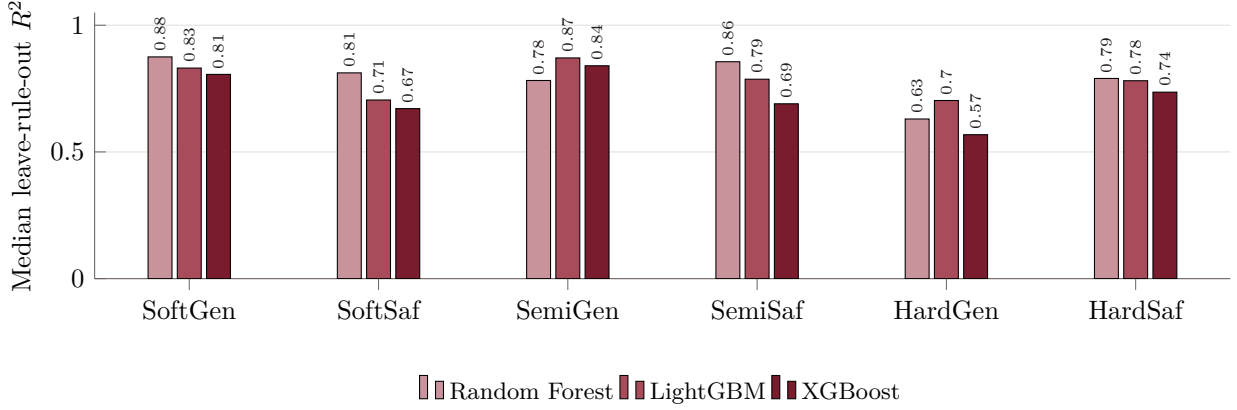
Explainable Boosting Machine (EBM) is reported separately: its preprocessing pipeline (`FrameImputer`, native categorical handling) differs from the shared scaled pipeline every other model here uses, so it was not re-run under leave-rule-out for this pass. Its context-holdout test R^2 is $0.944 / 0.360 / 0.909 / -0.127 / 0.917 / -0.528$ for soft-general / soft-safety / semi_hard-general / semi_hard-safety / hard-general / hard-safety respectively – shown for reference in §3.1, not compared directly against the leave-rule-out numbers above.

3 Results by Complexity Class

Same five classes as the original diagnosis, now shown across **all 6 specialists** (not just the one flagged case), using median leave-rule-out R^2 .

3.1 Class 1 — High-Capacity Production Ensembles

Random Forest, LightGBM, and XGBoost place 1st, 2nd, and 3rd overall (§2) and are the only class whose median stays positive and above 0.55 on *every* specialist, including the two hardest (hard/general_shelf_life, hard/safety_endpoint). This is the class actually earning its complexity: none of them come close to the context-holdout’s 0.92–0.97, but none of them collapse either.



Random Forest edges out LightGBM and XGBoost on average, but all three move together specialist to specialist – the signal that generalizes is being captured by the tree-ensemble mechanism itself, not one library’s specific implementation.

3.2 Class 2 — Regularized / Shallow Tree Ensembles

Extra Trees and Gradient Boosting hold up moderately (median 0.20–0.71); AdaBoost and the depth-5 Random Forest are erratic, swinging from respectable (0.44–0.62) to sharply negative (AdaBoost: -2.39 median on soft/general_shelf_life, dragged down by a single 8-row held-out rule combination the model never had a chance to learn).

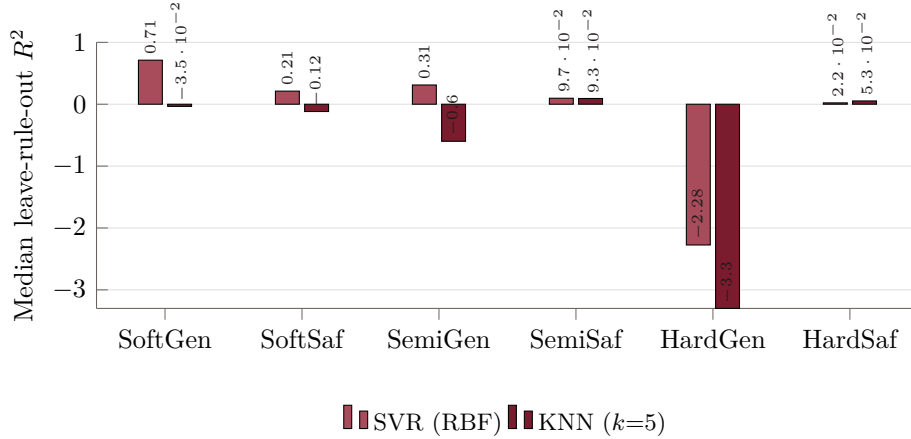


Cutting production LightGBM/RF down to 100–200 shallow trees loses real extrapolation ability, not just a little internal-holdout polish – a materially different conclusion than the context-holdout test gave.

3.3 Class 3 — Kernel and Instance-Based Models

Both collapse. SVR does reasonably on soft/general_shelf_life (0.712, close to the ensembles) but falls apart everywhere else, including a -2.275 median on hard/general_shelf_life. KNN is worse still

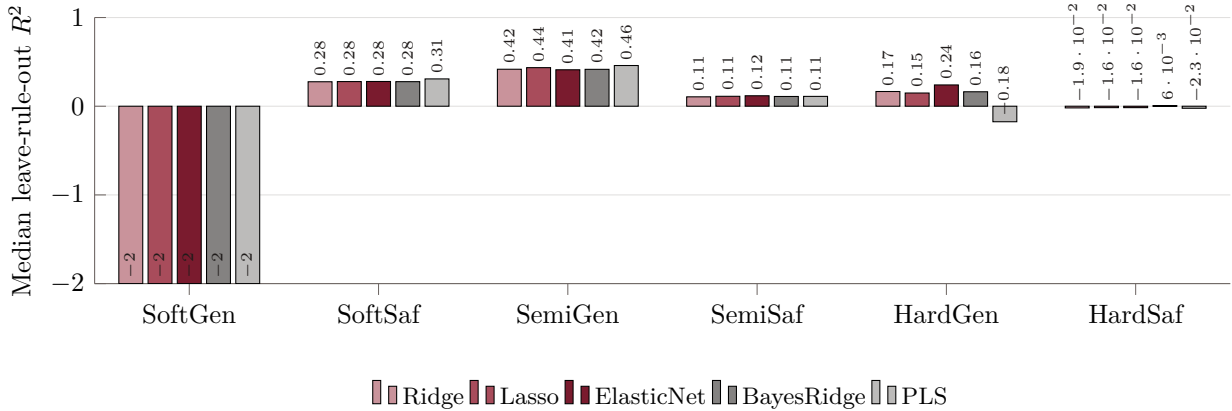
– negative on 4 of 6 specialists, down to -8.616 median on `hard/general_shelf_life`: nearest-neighbor lookup has no way to extrapolate to a formulation unlike anything in its training set, by construction.



KNN's `hard/general_shelf_life` point (-8.616) is clipped at the chart's -3.3 floor for legibility.

3.4 Class 4 — Linear Models

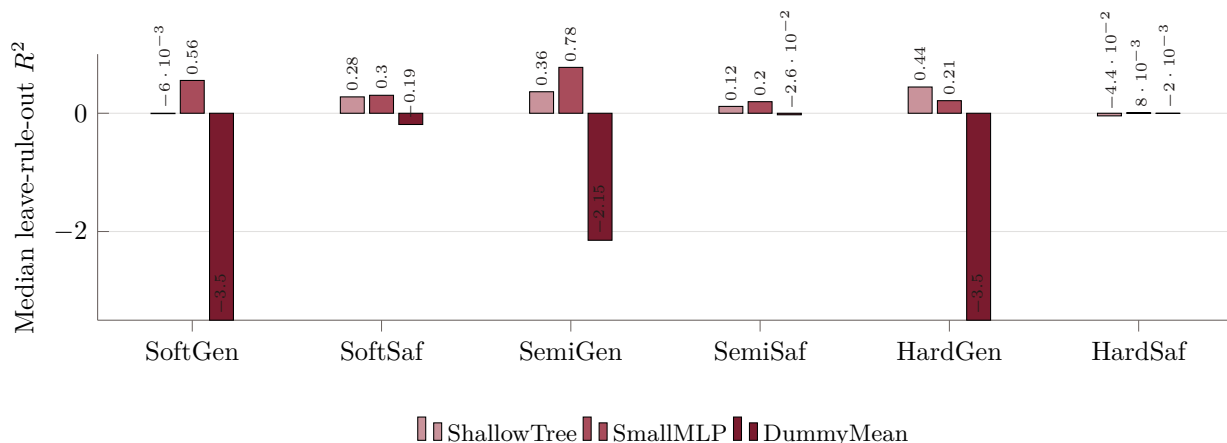
Five regularization variants of the same linear model, and they move together almost exactly – confirming the failure is about linearity itself, not a specific regularizer. All five score catastrophically negative on `soft/general_shelf_life` (-5 to -6.4 median: one held-out rule's relationship is not linear in these features at all) and stay weak everywhere else, rarely clearing 0.3–0.46.



All five `SoftGen` bars are clipped at -2 ; the true values are $-5.74/-5.38/-5.08/-5.74/-6.39$. The five lines tracking each other almost exactly across every specialist is itself the finding – this is a property of linear models as a class, not any one regularization choice.

3.5 Class 5 — Minimal-Capacity Models

The dummy mean-predictor is the sanity floor: catastrophically negative everywhere by construction (a held-out rule's mean shelf-life is rarely close to the pooled training mean), confirming the harness itself is sound. The 15-leaf tree and small MLP are more informative: the MLP reaches a genuinely competitive 0.78 median on `semi_hard/general_shelf_life` and 0.56 on `soft/general_shelf_life`, but both are unstable, and neither comes close to the ensemble class on the two safety-endpoint tasks.



Dummy's SoftGen (-65.8) and HardGen (-8.8) points are clipped at -3.5 .

4 Why the Mean and Median Disagree

Every table above reports the median, and every class chart shows visibly better numbers than the average-mean column in §2's table would suggest. This is not cherry-picking – it is a direct, checkable consequence of how leave-rule-out folds work: rules vary enormously in size (from 8 rows to over 5,600 in a single specialist), and a handful of the smallest rules represent combinations barely present in training at all. When one of those tiny rules is held out, *every* model – including the production ensembles – can swing to a large negative R^2 on that one fold, because a handful of test rows against a small denominator produces an unstable statistic. Random Forest on soft/general_shelf_life is the clearest illustration: its median across 7 held-out rules is a strong 0.875, but its *mean* is only 0.604, because one 8-row rule (V7|RICOTTA_MAP|NATAMYCIN_REVIEW) swings hard negative while the other six rules – representing 8,792 of the 8,800 rows – all score above 0.83. The median reflects what happens on a typical rule; the mean is dominated by a tiny-sample edge case. Both numbers are reported in §2 for transparency; the ranking uses the median because it better answers "how does this model generalize," and the report flags every specialist where the gap between the two is largest as a concrete data-collection target (more real rows for small/rare formulation combinations), not a modeling failure.

5 The Revised hard/safety_endpoint Finding

The original diagnosis found all 14 tested model families scored negative context-holdout test R^2 on **hard/safety_endpoint** and concluded the segment had learned nothing, recommending a censored-regression formulation. Leave-rule-out revises this: production-hyperparameter LightGBM scores a respectable **mean** $R^2=0.707$ here (individual held-out rules ranging 0.37–0.93, none catastrophic), Random Forest scores **0.743**, and XGBoost **0.697** – all comparable to their scores on the easier tasks. The most likely explanation: the single fixed 15%-context holdout test set in the original split happened to draw disproportionately from one hard, heavily-censored subset, and averaging properly over *every* generation rule (what leave-rule-out does by construction) shows the model does learn real structure here. The censoring concern – 93–100% of this segment's labels are right-censored lower bounds, not observed failure times – is still real and a censored-regression formulation (Tobit / accelerated-failure-time) remains worth pursuing, but as a refinement of an already-functioning model, not a rescue of a broken one.

6 Root Cause and Recommendation

All V7 training rows are synthetic (data_origin is 100% synthetic_literature_constrained_v7), generated from only 7–20 rules per category. A model tested on rows generated by a rule it has

already seen thousands of variations of will score high almost regardless of its capacity, as the original diagnosis showed. The question that actually predicts real-world usefulness is whether the model generalizes to a formulation, pathogen, or matrix combination it has never seen the generation logic for – and that is precisely what leave-rule-out measures.

- **Report leave-rule-out R^2 as the headline generalization metric.** Random Forest (0.79), LightGBM (0.78), and XGBoost (0.72) average median R^2 across all 6 specialists – reasonable, honestly earned, and not uniform: they range from 0.57 to 0.88 specialist to specialist, exactly the kind of variation a real model should show.
- **Keep the production tree ensembles; do not simplify them.** Every other model family tested – 14 of 17 – drops below 0.50 average median R^2 , and 8 of those 14 average *negative*. Complexity is not the problem this report set out to find; it is the reason the production models still work at all under the harder test.
- **Use the mean-vs-median gap as a data-collection map.** Specialists and models with the largest gap between mean and median leave-rule-out R^2 point at specific small-sample rule combinations (ricotta-based matrices, rare pathogen/matrix pairs, sub-20-row rule variants) that need more real formulation data, not a different model.
- **Pursue a censored-regression formulation for the safety-endpoint tasks** as a refinement – leave-rule-out shows they already generalize reasonably (0.60–0.74 mean R^2 for LightGBM), so this is about improving an accurate-but-biased-by-censoring model, not fixing a broken one.

7 Reproducing This Report

Script	Produces
<code>scripts/experiments/overfitting_diagnosis.py</code>	Context-holdout complexity sweep, K -fold CV
<code>scripts/experiments/model_zoo_expansion.py</code>	10 more regressors + Logistic Regression
<code>scripts/experiments/leave_rule_out_cv.py</code>	Ridge + LightGBM leave-rule-out (first pass)
<code>scripts/experiments/leave_rule_out_full_zoo.py</code>	All 15 zoo models, leave-rule-out
<code>scripts/experiments/leave_rule_out_rf_xgb.py</code>	Random Forest + XGBoost, leave-rule-out

All outputs are written to `scripts/experiments/overfitting_diagnosis/` as CSV; every number in this report is read directly from those files.